Lab Report # _____

CSC/EGR 4110 · Internet of Things

Name: _____ **Lab partner(s):** _____ **Date:** _____

Goal of today's lab, in your own words:

Check-off tests / drills: number from the lab page; instructor initials once you have shown it working

Test	Prediction (if asked)	What happened (numbers, counts, observations)	Pass?	Initials

MicroPython code: the lines you wrote or changed, not the whole file. File name: _____

```
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
```

What you measured or observed: real values from your own system, not copied from the lesson

What went wrong, how you found the cause, and how you fixed it (or where you got stuck):

Wiring and Parts

Lab # _____ **Name:** _____ **Date:** _____

Draw a line from each pin you use to a hole on the breadboard (or to a Pico pin). Write the Pico pin and any resistor value on the wire. Network labs: sketch the message flow in the margins.

SENSORS AND INPUTS

Push button A / B
or lever microswitch (bumper)
leg
leg

Reed switch + magnet
leg
leg

LDR light sensor
with 10 kΩ divider resistor
leg
leg

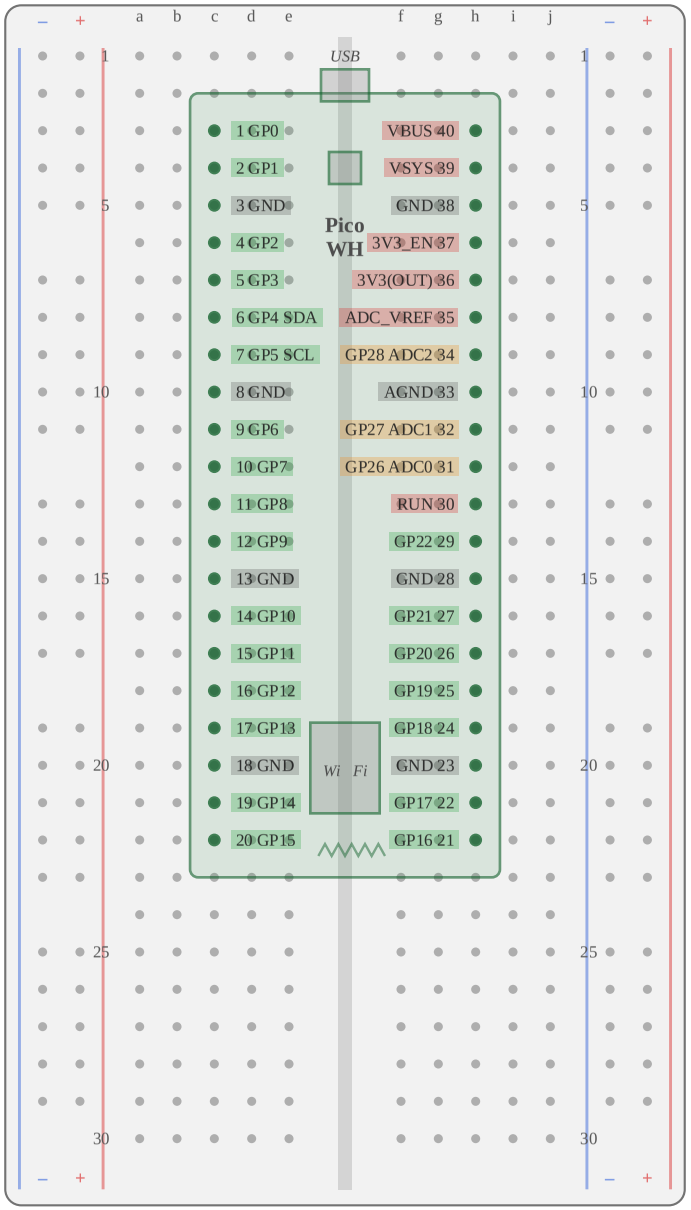
Thermistor
with 10 kΩ fixed resistor
leg
leg

Potentiometer (knob)
outer
wiper
outer

HC-SR04 ultrasonic
ECHO needs 1 kΩ + 2 kΩ divider
VCC
TRIG
ECHO
GND

MPU6050 IMU
I²C
VCC
GND
SCL
SDA

LCD1602 + I²C backpack
GND
VCC
SDA
SCL



GND Power GPIO ADC

Pico sits across the channel, pins in columns c and h.

OUTPUTS, POWER AND PASSIVES

Red LED
needs a 220 Ω resistor
anode (+, long)
cathode (–)

Green LED
needs a 220 Ω resistor
anode (+, long)
cathode (–)

Resistors
220 Ω, 1 kΩ, 2 kΩ, 10 kΩ
end
end

Servo (SG90)
own battery pack, never 3V3
signal (orange)
V+ (red)
GND (brown)

TA6586 motor driver + motor
IN1
IN2
VM (battery +)
GND
M

4 × AA battery pack
join its ground to the Pico GND
+
–

Capacitor 470 µF
across the servo power rail
+ (long)
– (stripe)